

MIDI Tool Box

An Interactive System for Music Composition

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the requirements of the degree of
Doctor of Philosophy

by

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**"Of course, you realise I have an advantage,
because I don't often call myself a composer."**

David Tudor, Source, Issue 6, p.8.

Prologue Rationale for a purpose-built musical instrument design

Throughout the history of western music, composers have frequently been proactive in the design of musical instruments. Rapid developments in electronic technology have radically transformed the focus of many composers and as the C20th draws to a close, an increasing number have become involved in making purpose-built instruments as an extension of their compositional craft. An attempt to survey major developments in computer music from this perspective is made in Chapter 1 : The emerging tradition of composer/instrument-builders.

I have used the term **composer/instrument-builders** to mean any composers who have "got their hands dirty" in the process of musical instrument development. This term is useful to differentiate between composers who simply write music - or provide musical content - for a new instrument and those actively involved in the development of a new instrument or system.

Using the term composer/instrument-builder highlights a musical change that has taken place in the C20th. This change is not adequately addressed by the discourse of contemporary musicology. Musicological discourse tends to focus on scored works as these lend themselves more easily to analysis by conventional musicological methods. It tends to overlook the work of many composers, whose hands-on involvement with technology, has contributed to the musical change being made in contemporary culture. In doing so, it has failed to provide a comprehensive view of music in the C20th. Alternative musicological discourse is needed to include the rediscovery of the role of improvisation in the creation of music and the emergence of alternative modes of musical interaction brought about by new technology.

The musical instrument design project described in the main body of this thesis has come about as a result of my work as composer, which in turn has been stimulated by my continuing involvement in the development of new technology. Two algorithmic compositions were produced as an extension of my earlier work in composition using purpose-built analogue and digital hardware. Using hardware which I had modified in 1984 to investigate a serial communications protocol known as MIDI, these two works - *Monophonic Variations* (1986)

and *Polyphonic Variations* (1988) - were created to control commercial MIDI systems interactively. A new design known as the **MIDI Tool Box** has evolved from these works which are analysed in Chapter 2 : Algorithmic compositions leading to the MIDI Tool Box.

This instrument has now provided a technological means that has allowed myself and others to create diverse musical applications. These are covered in Chapter 3 : MIDI Tool Box : hardware and musical applications. The open-ended nature of the tools provided has placed myself and other musicians in a position to define the musical starting point of each application. Each user may radically redefine the user interface.

The process of developing my own applications for the MIDI Tool Box has revealed both the limitations in the micro-controller-based instrument's processing capability and my personal dissatisfaction with the fairly crude form of data reduction offered by the MIDI protocol itself.

Unlike playing a traditional bowed or wind instrument, playing a MIDI software instrument primarily involves triggering discrete sounds in ways which do not necessarily involve a player in continuous control over the production of sound for the duration of the event, such control being a desirable form of musical interaction. Musicians generally recognise a difference between working with discrete triggered events and working with continuously controlled sounds, a difference akin to playing with modular shapes like lego blocks as opposed to playing with malleable material like plasticine.

This approach to creating new instrumental sound required the MIDI Tool Box to acquire a signal processing capability for the production of audio together with an improved protocol for communication between other musical instruments. Chapter 4 : A design review looks at some of the signal processing options available. An ASIC chip known as the A4 further enhanced the MIDI Tool Box design allowing it to process audio signals. A new protocol for communication between musical instruments is investigated.

In Chapter 5 : A4 MIDI Tool Box : hardware prototype, tests & application describes development of the new signal processing hardware system and explains some of the procedures developed to test it. The implementation of the first live musical performance application using the A4 MIDI Tool Box is described. In this musical application, I have intentionally established a musical link between modern signal processing technology and one of the world's longest surviving traditions of improvised music viz., Carnatic (south Indian classical) music. This is a tradition which has been propagated over several millenia without the use of notated scores. It is also a tradition in which there have been instances of adaptation and appropriation of technology by the musicians themselves. In the first performances of the first musical application of the new design, Carnatic musicians perform together with the A4 MIDI Tool Box.

In Chapter 6 : The MIDI Tool Box and beyond the author reviews the musical possibilities made possible through involvement in the design process. The author evaluates his work against the changing technological landscape in which trails left by other composer/instrument-builders can now be more easily identified. Projections are made about future developments based on networked musical instruments and their potential musical applications.

In the sleeve of this volume, there are two CDs. The first of these contains the two musical works which led to the design of the MIDI Tool Box together with subsequent works created by the author using both the MIDI Tool Box and the A4 MIDI Tool Box. The second of two CDs contains mostly software listings which are included for reference and to make this software available in the public domain. It also contains "hot link" versions of various URLs found in the end of this thesis as well as information obtained from some of these links on a specific date.

Chapter 1 The emerging tradition of composer/instrument-builders

Throughout the history of western music, composers have frequently been proactive in musical instrument design. The modern tradition of experimental composition began with composers whose purpose-built instruments, though non-electronic, pre-empt the most recent developments in music technology. The 1960's saw the emergence of a handful of composer/instrument-builders who performed on as well as designed the electronic instrument or system for which they composed. These composers paved the way for a new generation spurred by the commercial availability of analogue electronics in the late 1960's and the advent of the microprocessor in the mid 1970's. As computer science developed, some composers began to author their own composition software while others even pioneered the design of algorithms for computer-generated sound. Since the arrival of MIDI in 1983, when it became possible for large numbers of musicians to accept the use of computers for performing music, music technology has continued to develop and the level of involvement of musicians in the development process continues to grow. New technology has enabled musicians to cross the boundaries between disciplines in a way that differentiates the composers/instrument-builders in this epoch from composers whose names have been associated with new instruments in previous epochs.

Technological change in the C20th has had an enormous impact on the social infrastructure of music production. Recording technology has given listeners unprecedented power to exercise choice about their musical preference thereby affecting the livelihood of musicians. Even with government and private sponsorship, large orchestras and opera companies have barely managed to survive in the C20th and fewer composers in the future will look to such institutions for commissions to sustain their livelihood. The majority of composers in the C20th have looked to music for film, television and radio, or other channels for the dissemination of popular music as their primary source of income. New technology has become more accessible throughout the C20th. A growing number of composers are now involved in musical instrument design. By being involved in this way, these composers have availed themselves of opportunities to steer technological development in ways that shape the means by which music is produced. It is the work of these composer/instrument-builders that is the focus of this chapter.

In tracing the role of composer/instrument-builders in the development of musical instrument technology, this discussion places development in five main categories.

- Pre-electronic Purpose-built Instruments
- Purpose-built Live-electronic Instruments Pre-70's
- Software Instruments
- Composer-authored Composition Software
- Purpose-built Live-electronic Instruments & Systems Post-70's

1 **Classification of Composer/instrument-builders**

In order to provide the context for the work described in the remainder of this thesis, the term **composer/instrument-builders** is used to mean any composers who have "got their hands dirty" in the process of musical instrument development, as previously stated (see Prologue, p. xviii). The focus of the survey is on composer/instrument-builders whose musical output has been dependent on the development of some new electronic technology and whose initiative and creative energy has been redirected towards making this possible.

Not every composer/instrument-builder is represented, nor is it possible to do so. The scope of this survey has largely been limited to those who have built their own electronic hardware, or devised a new programming technique - as opposed to composers who have written music (in the conventional sense) for new electronic instruments. However, not every composer included here has contributed directly - as a result of their excursions outside composition - to the current state of music technology; some are simply representative of new musical trends that have since emerged.

C20th composers who built non-electronic instruments have been excluded from this survey ; their inclusion would be more appropriate in another context ¹. The many composers who have worked extensively in electronic tape studios have also been excluded ² ; their work deserves a place in the history of the development of recording technology and its effect on composition and performance. As a composer who has designed a new instrument, my primary concern in this thesis is with the role of technology in the evolution of new instruments which bring about new modes of performance.

An initial attempt was made to devise categories chosen on the basis of the different technologies used, ie. pre-electronic, analogue electronics, digital hardware, computer software, MIDI, etc. However, it soon became clear that the work of many composer/instrument-builders confuses categories devised in this way. The set of categories finally selected is based on areas of related musical concern, as this allows the broadest range of instrument-building activity among composers to be classified.

As work on this survey progressed, it became clear that the number of composer/instrument-builders represented here is but the tip of an iceberg. In another context, this survey would also include the work of many other composers ³. Their inclusion would only add further weight to the argument advanced by the author in this chapter. Its omission from this survey is not an oversight but one that was ultimately imposed by constraints of space and deadlines.

¹perhaps surveyed from the context of ethnomusicology, improvisation and composition in C20th music.

² such as Karlheinz Stockhausen, Pierre Schaefer, Pierre Henri, Luciano Berio, Luigi Nono, Mario Davidovski, Henri Pousseur, Morton Subotnick, Krystof Penderecki, Ilhan Mimaroglu

³ composers like James Tenney, Alan Lamb, Martin Bartlett, Tom Erbe, Mamoru Fujieda, James McCartney, Jon Miselle, Nam June Paik, Iain Mott, Carter Scholz, Trimpin, Ed Kobrin, Laurie Spiegel, Kimmo Vennonen, Kazeo Uehara, and more

The selection criteria for those represented within each category was particularly problematic. For example, a category is devoted to the work of some of the earliest C20th composers, even though, except for one, they did not build electronic instruments. At the same time, not every composer who built non-electronic instruments could be included in the first section of this overview. It was necessary to include this category so that undue importance was not placed on some later electronic musical instrument development. For the same reason, a category is devoted to those who built analogue systems. And even though some of the early developments in computer music have more in common with recording technology than performance technology, inclusion of a section devoted entirely to software instruments - and another to composer-designed software designed to play them - is necessary in order to understand the role of the computer in the development of modern performance instruments. For each new category, a different selection criteria has been specified, stating how each of the selected composer/instrument-builders is representative of the category in which their work has been cited. Where appropriate, some composer/instrument-builders are included under more than one category.

The categories attempt to provide a comprehensive overview of C20th composer/instrument-builders by including a broader range of activity than those areas of specific interest to the electronic design described in the main body of this thesis. Some composer/instrument-builders reviewed in this chapter have introduced musical techniques which are referenced in later chapters with detailed explanation. This serves to identify the similarities between the work of others and the author's work and provide a better understanding of its context.

Information on composer/instrument-builders reviewed in this chapter is taken not only from journal articles, but also from reviews of both recordings and concerts. Liner notes such as those on recordings released by Artifact Records ⁴ & Lovely Records ⁵ have provided a useful source - and in some cases sometimes the only source - of written biographical and musical information. Web sites have also been used.

1.1 Pre-electronic Purpose-built Instruments

A tradition of instrument-building existed among composers before computers were used in music. Some of the first composers who built new instruments include Percy Grainger, Harry Partch, Ivor Darreg, Conlon Nancarrow, John Cage, George Antheil. These composers identified a musical impasse brought about through the dominance of previous technology viz. the restriction of equal tempered tuning combined with the limitations of the conventional music keyboard. Grainger (1882 - 1961), Partch (1901 - 1974) and Darreg (1921 - 92) invented an entirely new set of instruments to realise new systems of tuning while Nancarrow (1912 - 1996), Cage (1912 - 1992) and Antheil (1900 - 1959) radically extended the ability of existing instruments in order to realise new sound possibilities or new modes of performance.

⁴ Artifact Records, 1998: biographical information on Bischoff, Brown, Horton, Perkis, Polansky, Trayle, The Hub obtained from CD liner notes at <http://artifact.com/>

⁵ Lovely Records : biographical information on Behrman, Bischoff, Cage, Chadabe, Collins, Harmonic, Kuivila, Lewis, Lucier, Mumma, Neill, Rosenboom, Tudor at <http://lovely.com/artists/browse-a.html>

1.1.1

Grainger

Before the turn of the century, Grainger recognised the arbitrary nature of a tuning system which quantised each octave of the audio spectrum into 12 equal-tempered divisions. Like his contemporaries ⁶, Grainger too was involved in the redesign of piano keyboards ⁷ in some of the early attempts to find alternatives to the harmonic language that had grown out of 12-note equal-tempered tuning. Grainger collaborated with Burnett Cross between the 1930s to the 1950s, to construct mechanical instruments which he called "free music" machines ⁸. These were pneumatically sequenced devices which produced gliding tones.

[Bird, 1982; Burt, 1991; Cross, 1989; Grainger & Cross, 1996; (CD)]

1.1.2

Partch

Just-intonation tuning was a primary motivation for Partch's purpose-built instruments - some 23 in all. These were designed purely for realising compositions written in his 43-tone per octave tuning system. In just-intonation, an interval between two pitches is the distance between points on the natural harmonic series. When this is expressed as a fraction, the higher pitch is the numerator, the lower is the denominator. Partch's harmonic system classified intervals into groups according to the odd-numbered identities - or limits - present in the numbers which define the interval. This system is described in his monumental tome on tuning ⁹. After experimenting with tunings used

⁶ Busoni, Vishnegradsky, Haba and Ives

⁷ Grainger, & Cross [undated -1952]: Butterfly Piano

⁸ Grainger, & Cross (undated -1952): , Reed-box Tone-tool, Hills & Dale cardboard roll, Kangaroo Pouch machine

⁹ Partch, H [1974]: Genesis of Music da Capo Press

by the ancient Greeks whose scales were mostly tuned in 3-limit and 7-limit just intonation, Partch devised a 43-tone scale based on 11-limit just intonation. He demonstrated the necessity to build an entire set of instruments in order to create a sound world that could not be explored using existing musical instrument technology. The sound world of just-intonation which he rediscovered through his unique instruments and compositions, has gradually become more accessible in the mainstream of music since the advent of MIDI-driven microtonally tuneable synthesisers.

[Partch, 1991; Partch, 1974; Drummond, 1994; Partch, 1940s: (CD); Partch & Cage, 1940s: (CD)]

1.1.3 Darreg

Partch's concern with non-equal tempered tuning was matched by Darreg's exploration of equal divisions of the octave other than 12. He designed and built purpose-built instruments - some of these electronic - and through them explored a vast array of microtonal tuning systems. Darreg composed for and improvised on the many instruments he built ¹⁰. He made extensive charts for comparing the effect of various equal divisions of the octave other than 12 on the listener's mood.

[Darreg, 1993; Darreg, 1986; Darreg, 1974; Varga, 1994; Darreg, 1953-1992: (CD)]

¹⁰ Darreg, I [1953-1992]: Detwelvevulate CD Ivor Darreg Memorial Fund ID100 including the Electronic Keyboard Oboe in 1956 one of the first synthesisers ever built, several years before Moog's and Buchla's first voltage-control synthesisers

1.1.5

Nancarrow

Nancarrow adapted the player-piano to create rather than reproduce music. The bulk of Nancarrow's musical output ¹¹ was created by systematically punching holes in paper rolls for specially modified player pianos. In so doing, he explored rhythmic complexity in ways that were hitherto unimaginable without mechanically sequenced playing. Effectively without access to any means to have his music played by others - because of political ostracisation - Nancarrow bypassed the problems of realisation in performance working closely with an instrument that became a virtual performer. Sequences could be reprogrammed, if he so desired, by gluing paper over punched holes in the player piano rolls. Nancarrow's composition was no longer confined by the inability of conventional notation to articulate his rhythmic concepts. Long before MIDI technology gave personal computer owners access to means to compose and hear unplayable musical sequences, he demonstrated the independence from notational systems that is possible using automated sequencing.

[Place, 1983; Nancarrow, 1950-1984; (CD)]

1.1.6

Cage

Cage was one of the first composers to broaden the range of sound timbres used in music by developing new instruments which incorporated non-musical sounds especially for percussion, to create music of immense rhythmic and timbral variety. The most famous of these was the **prepared piano** which extended the range of timbral possibilities of the conventional piano by the insertion assorted objects between the

¹¹ Nancarrow, 1950-84: Studies for Player Piano CD Wergo

strings. Extending the Futurist ¹² aesthetic, Cage's earliest manifesto involved "producing sounds by any means possible ... " ¹³. This aesthetic towards new sounds ¹⁴ became important for a new generation of musicians that emerged when sampling technology became commercially available after the mid 70's. Central to this aesthetic was suspension of musical value judgements in order to discover unintended musical relationships that can be heard in every acoustic environment.

[Cage & Tudor, 1960-88: (CD); Cage, 1939-51: (CD); Cage, 1944-45: (LP); Partch & Cage 1940s: (CD)]

1.1.7 Antheil

Antheil, for his composition *Ballet Mecanique*, devised a coding technique for synchronising multiple player pianos which was later to become the basis of a successful US patent for frequency hopping ¹⁵ and which was employed to prevent jamming of radio-frequencies used to control weapons guidance systems ¹⁶. In more recent times, frequency hopping has become the basis of modern spread-spectrum communication ¹⁷. The synchronisation technique was like a mechanical implementation

¹² Luigi Russolo & Antonio Russolo the first composers to devise instruments which use noises of the modern world as a musical resource

¹³ J.Cage : Silence Wesleyan University Press 1961

¹⁴ in his later works for magnetic tape, radios, phonograph records and other media.

¹⁵ H.K.Markey & G.Antheil : Secret Communication System US Patent 2 292 387 Aug.11 1942 filed June 10 1941

¹⁶ "The system design took special advantage of the composer's know-how to synchronise the radio transmission and reception frequencies by means of twin, identically crypto-code slotted, paper music rolls like those used in player piano, audio-frequency mechanisms. Indeed, Antheil had already achieved such synchronisation precisely in his multi-player-piano opus of the 1920's, *Ballet Mechanique* [G.Antheil : Bad Boy of Music p.185 Doubleday NY 1945]. Their invention disclosure points out that an FH (frequency-hopping) repertoire of 88 radio frequencies could readily be accommodated." from Robert Price : Further Notes and Anecdotes on Spread-Spectrum Origins pp.89-91 *IV. Shortly before Pearl Harbour: The Lamarr-Antheil Frequency-Hopping Invention* IEEE Transactions on Communications, p.90, Vol. COM-31, No.1, Jan. 1983.

¹⁷Patent document was downloaded from <http://www.ncafe.com/chris/patent/index.html>

of communication between musical instruments that can now take place over synchronous networks ¹⁸.

[Antheil, 1925; Antheil & Markey, 1941; Price, 1983; Antheil, 1925: (CD)]

Though many of these instruments are non-electronic, their inclusion here is justified by recent developments. One such development, anticipated by Grainger, Partch and Darreg, is experimental tuning. These composer/instrument-builders sought tuning that was not derived from the piano keyboard, unlike the other early C20th composers ¹⁹ who experimented with tunings that involved further subdivision of the equally tempered intervals of the piano. It was Partch who rediscovered the virtues of just intonation and devised instruments based on a single system of quantised pitches in order to realise his ideas; Darreg further experimented with multiple systems of quantised pitches; and Grainger's ideal pitch system was not even quantised but continuous. The dominance of the piano as a composition interface was challenged in quite a different way by Cage - whose prepared piano realised other possibilities for musical timbres using conventional keyboard technique and notation - and Nancarrow - whose player piano realised new possibilities for rhythmic invention independent of live performance. Antheil's synchronisation of multiple player pianos was not simply an extreme form of mechanised 19th century musical instrument technology but a foretaste of musical instrument networking that is now possible in the era of digital network technology.

¹⁸ in Sept. 1997, Antheil & Markey were awarded medals from the Electronic Frontier Foundation for their services to the field of spread spectrum communication (<http://www.ncafe.com/chris/patent/index.html>)

¹⁹ Busoni, Vishnegradsky, Carrillo, Haba and Ives

1.2 Purpose-built Live-electronic Instruments Pre-70's

Experimental musicians of the sixties who built electronic instruments provide an important link between the early composer/instrument-builders (outlined in 1.1) and later composer/instrument-builders who have developed purpose-built electronic systems. Their inclusion also highlights one of the major paradigm shifts that has taken place in music of the twentieth century, namely the transition from technology based on paper to technology based on silicon. The transition was a gradual process which took place in parallel with new developments in consumer electronics brought about by the transition from valve to semiconductor technology. Audio and broadcast radio equipment became more portable and instantaneous giving rise to the proliferation of consumer hi-fi products and cheaper components. By comparison with valve technology, hands-on experimentation using low voltage semiconductor circuitry was also safer. As a result, musicians with little background knowledge in electronics and no longer content to be the passive recipients of technological innovation, could become involved in building their own electronic instruments.

Source Magazine published the Circuit-book in one of its issues ²⁰ in 1969. This was a collection of circuit schematics developed and used by musicians. It gave artistic endorsement to hands-on experimentation with electronics. Source was succeeded by Electro-notes ²¹ a periodical devoted to electronic music circuit design which continued publication from 1972 until the early 80s. These publications helped give musicians a practical understanding of

²⁰ Source : Music of the Avant-garde, 1969: The Circuit-book Issue No. 9, pp. 51-59

²¹ Electronotes 1972 to 1982 publ. Ithaca University, Ed. Bernard Hutchins

technology that had hitherto been idealised by commentators such as Norbert Wiener ²² and Marshall MacLuhan ²³.

A representative list of composers who built systems using analogue electronics include :

David Behrman, Max Neuhaus, David Tudor, Alvin Lucier, and Gordon Mumma

1.2.1 Tudor

Tudor, pianist-composer, and life-long collaborator of Cage, created some of the first purpose-built circuits for electroacoustic music in the early 1950s. In *Rain Forest* (1974) audio signals are fed into resonating objects made from assorted types of material. Transducers are attached to the resonating objects (which are found rather than manufactured) and which become loud-speakers. Feed-back is produced and controlled both by the frequency characteristics of the material and by voltage-control filter circuits designed by Tudor. Tudor never abandoned this close association with hardware development. A new synthesiser design using Intel's neural network chip is based Tudor's early live electronic performance techniques ²⁴. Tudor ²⁵ created a series of works for the new instrument ²⁶ starting with *Neural Network Plus* (1992) ²⁷, and ending shortly before his death with *Neural Synthesis Nos.4-6*.

²² Weiner, N. [1988]

²³ McLuhan, M [1965]

²⁴ Warthman, 1997: *The Neural Network Synthesiser* at <http://wolfpack.bard.edu/tudor/index.htm>, Jan.1998

²⁵ in collaboration with Warthman, Holler & Thorson

²⁶ Tudor home page at <http://wolfpack.bard.edu/tudor/index.htm>; Warthman, 1997: (article with instrument photograph) at <http://www.emf.org/tudor/Articles/warthman.html>

²⁷ for the Merce Cunningham Dance Company and the Paris Opera

[Warthman, 1997; Cage & Tudor, 1960-88: (CD); Tudor, 1972-81: (CD); Tudor, 1996: (CD)]

1.2.2 Behrman

Behrman's earlier work *Wave-train* (1968) used purpose-built analogue circuits to produce feed-back through the strings and frame of a piano. He interacts with this throughout the performance by tuning the feed-back. Interactivity of a different kind is a feature of his most recent work in which computers and purpose-built electronics are used as part of an installation. *Keys to Your Music* (1989), an example of his most recent work, is an interactive sound exhibit involves modified kalimbas interfaced through purpose-built circuitry to a personal computer. This type of interactivity gives a hands-on music-making experience to people in the lobby of a gallery or museum.

[Behrman, 1991; Dodge & Jerse, 1985; Garvey, 1981; Behrman, 1968; Behrman, 1991: (CD); Behrman, 1977: (CD)]

1.2.3 Neuhaus

Neuhaus, a distinguished percussionist who once played the standard post-war modernist repertoire, was also a pioneer in creating interactive sound installations. His *Telephone Access* (1968), created during an artist-in-residency program at Bell Telephone Labs, was a interactive installation to which the public gained access via an advertised telephone number ²⁸ and is described as "*a piece of music in the form of an electronic system*" ²⁹. The circuit diagram is published in Cage's book *Notations* ³⁰.

²⁸ Sept.4- Oct.14 1968, Neuhaus, M. [1968]: *Telephone Access*

²⁹ p.56 *ibid*

³⁰ in Cage, 1969: *Notations*, "Neuhaus, *Max-feed*, 1 page, 45.5cm x 30.2 cm., transfer symbols on acetate"

Public Supply is a radio broadcast where the public phone-in and their sounds are broadcast. Neuhaus' *Water-whistle* (1970) installation, though non-electronic, is a interactive swimming pool in which the sounds are only heard underwater, pre-empting the underwater electronic installations of Redolphi ³¹. Sound installations have become an increasingly important presentation format used in museums and exhibitions. Neuhaus' work is primarily in this medium.

[Redolphi, M.1981-94; Neuhaus, 1970; Neuhaus, 1968]

1.2.4 Lucier

Lucier has made many connections between the world of music and other physical systems. *Music for Solo Performer* (1965) was the first work to use a performer's alpha rhythms as a control mechanism for activating assorted percussion instruments. *Music for Pure Waves, Bass Drum and Acoustic Pendulums* investigates wave propagation through transducers that produce rhythmic motion in a pendulum that plays a drum. *Spherics* (1981) is a musical investigation of the same celestial phenomena that instigated the birth of modern radio-astronomy ³². Lucier's use of stretched wires as acoustic wave-guides in *Music on a Long Thin Wire* (1972) predates recent musical research into digital wave-guide synthesis³³.

[Lucier, 1980: (CD); Lucier, 1969: (CD); Lucier, 1988: (LP); Jansky, 1935; Lucier, 1970; Lucier, 1965: (LP)]

³¹ Redolphi, M [1981-94]: *Sonic Waters* and *In Corpus* underwater interactive environments at <http://www.imaginet.fr/manca/music/biouk/mrbioguk.html>

³² Jansky, K. [1935]: A note on the source of interstellar interference Proceedings of IRE, Vol. 23 p. 1158

³³ Smith, J. [1992]

1.2.5

Mumma

Political motivation and purpose-built electronics feature in many of Mumma's large theatre works such as *The Dresden Interleaf 13 February 1945* (1965), *Megaton for Wm. Burroughs* (1963), *Music from the Venezia Space Theatre* (1964). *Bandoneon* (1970) is a smaller theatrical work in which the live sound of a "bugged" piano-accordion is processed electronically. The accordion is played on stage by a puppet-like performer who is remote controlled by strings operated offstage by an unseen performer. Mumma built electronics used in works like *Mesa* ³⁴ and *TV Re-Run* (1974) which he created for the Merce Cunningham Dance Company. In the second piece music is created by dancers each wearing a belt to which pairs of FM radio-transmitters are attached. Transistor radio sets receive signals transmitted by the dancers whose positions control sound generators heard in the performance. Mumma, with Tudor, were responsible for realising the theremin-based live-electronics activated by dance movement in Cage's *Variations IV* (1965).

[Mumma, 1979, Cage, 1969]

³⁴ in Cage, 1969: Notations, "Mumma, *Mesa*, 2 pages, 27.5cm x 21.5 cm., ink on graph paper"

These composer/instrument-builders performed on as well as designed the electronic instrument or systems for which they composed. Their analogue electronic systems brought about new possibilities for musical interaction. The concept of feed-back was no longer just a property of the analogue circuitry used but took on musical significance. It involved not only the electrical paths between the outputs and inputs of a system, but the larger notion of feed-back between the performers of an ensemble, or between the creators of a musical event and its audience. These composer/instrument-builders created a new form of real-time composition in which the composer takes on the roles of instrument builder and performer. They frequently created new ways for the listener to affect, as well as be affected by the musical environment made through their instruments.

1.3 Software Instruments

Just as analogue electronics radicalised the concept of musical instrument design, the development of digital electronics introduced unprecedented flexibility into the design of musical instruments. At first, computer music was a novelty somewhat like the gramophone. It was associated with passive listening. Like radio-astronomy, computer music technology began at Bell Telephone Laboratories as part of a research effort in telecommunications ³⁵. From the first computer music composition languages written by Max Matthews, there evolved two classes of software :

- the instrument ie. the sound synthesis algorithm and
- the score ie. the sound sequencing algorithm.

³⁵ under the direction of Max Matthews and John Pierce

The work at Bell Labs involved composers ³⁶ and the collaboration that took place there between musicians and computer scientists was later perpetuated by composers at Universities like Stanford ³⁷, Princeton ³⁸, Illinois ³⁹, and MIT ⁴⁰.

Similar institutions sprang up in other places in the US and Europe, eg. at the Institute of Sonology, The Hague ⁴¹ in the late 1960s and in Paris where two research institutes were established, one for mathematics and music, CEMAMu ⁴², in 1974 and another purely for dedicated computer and acoustic music research, IRCAM ⁴³, in 1976.

Although most of the initial compositional development at Bell Labs used a language developed by Matthews called Music3, it was **Music4** which extended the development of computer music beyond Bell Labs. Based on Fortran, Music4 was ported to computer systems at a number of universities and quickly produced as many variants as there were systems in use. Composers learned to accept the tedium of adapting code to accommodate differences in the available hardware and soon became fluent in developing new software of their own. Other languages sprang up, all derived from Music4. The enduring feature of these languages was the two-fold division of composition software as orchestra and score.

³⁶ such as John Chowning and Jean-Claude Risset, and later Richard Moore, James Tenny, Laurie Spiegel and Emmanuel Ghent

³⁷ John Chowning, Lorn Rush, John Grey & James. A. Moorer

³⁸ Hubert Howe, Milton Babbitt, Otto Leuning, Vladimir Ussachevsky, Paul Lansky

³⁹ Hiller, Brun & Martirano

⁴⁰ Vercoe & Dodge at Massachusetts Institute of Technology

⁴¹ Koenig & Laske

⁴² acronym for Centre d'Études de Mathématique et Automatique Musicales directed by Xenakis

⁴³ acronym for Institut de Recherche et Coordination Acoustique/Musique set up under the direction of Boulez

By the end of the 60s some composers had acquired sufficient experience to develop sound synthesis algorithms. For the first time, composers were able to design software instruments. They devised many of the new synthesis techniques now used by other composers. Without their efforts software instruments that are so important today in the realisation of music through computers would simply not exist.

An exhaustive list of composers involved in the design of software instruments is beyond the scope of this chapter. A representative list of composers who have designed software instruments is :

Jean-Baptiste Barriere, John Chowning, David Jaffe, Paul Lansky, Richard Moore, Jean-Claude Risset, Xavier Rodet, Barry Truax.

1.3.1 Risset

Probably the most important composer in the early history of computer music was Risset. The many software instruments he designed have since been widely used by other composers of electro-acoustic works. With a background in engineering and acoustics, Risset's compositions are among the earliest computer music produced at Bell Telephone Labs. His software instruments ⁴⁴ include snare-drum, gong-like sounds based on additive synthesis techniques, one based on Shepard tones ⁴⁵, other techniques for producing inharmonic bell tones and harmonic glissandi. Many of these software

⁴⁴ documented in Introductory Catalogue of Computer Synthesised Sounds, Bell Labs. NJ. 1969; many of these can be heard in their Csound implementation by R. Pinkston at <http://www.vanderbilt.edu/Blair/Courses/MUSIC216/Rissett.html>

⁴⁵ a technique to produce continuous glissandi

instruments were designed for Risset's own compositions most notably *Mutations I* and *Inharmoniques*.

[[Pinkston, 1994](#); [Rowe, 1992](#); [Risset, 1991](#); [Risset, 1990](#); [Rowe, 1988](#); [Risset, 1985](#); [Risset, 1969](#); [Risset, 1966](#); [Risset, 1969](#); [Risset, 1969-85](#): (CD)]

1.3.2 Chowning

Chowning's FM synthesis technique allows various complex audio spectra to be generated by specifying just a few frequency ratios and modulation indices. It was first developed on a PDP-6 at CCRMA⁴⁶, Stanford for his composition *Sabelithe* (1966) as a means of making gradual transformations from one type of sound to another. The technique became widely used when it was licensed to Yamaha in the late '70s and marketed as the synthesis engine of the DX-7 series of keyboards. Commercialisation of Chowning's idea illustrates how research in computer music has directly affected the music industry. Another technique also devised by Chowning is concerned with audio spatialisation, used in his 1972 composition *Turenas*.

[[Green, 1988](#); [Chowning, 1973](#); [Chowning, 1971](#); [Chowning, 1966-81](#): (CD)]

1.3.3 Lansky

The linear prediction algorithm developed by Lansky is an extremely versatile method of manipulating pitch and modifying the timbre. Filter techniques devised by the composer allow the frequency characteristics of sampled speech and other sounds with noise components to be altered. This technique is used extensively in his *Six Fantasies on a Poem by Thomas Campion* (1979), *Idle Chatter* (1983) and *Guy's Harp* (1985).

⁴⁶ acronym for Centre for Computer Research in Music and Acoustics

Lansky's EIN scratchpad software is the most recent extension of this involvement with sound filtering techniques.

[Lansky & Steiglitz, 1995; Warren, 1993; Belet, 1993; Rowe, 1990; Dodge & Jerse, 1985; Roads, 1983; Lansky & Steiglitz, 1981; Lansky, 1997: (CD); Lansky, 1994: (CDs); Lansky, 1992: (CD); Lansky, 1990: (CD); Lansky, 1982: (LP)]

1.3.4 **Barriere & Rodet**

Composers Barriere and Rodet, in collaboration with Potard are the authors of Chant. The synthesis model *"uses a scheme that creates a spectrum containing resonances without the use of filters. The system that implements the scheme is named "Chant". Rodet's method, which he calls FOF (fonctions d'onde formantique) synthesis, simulates the speech by triggering multiple oscillators in a harmonic series for short bursts at the beginning of each fundamental pitch period. The user specifies the centre frequency, bandwidth, and amplitude of the five formants, as well as the fundamental frequency, amplitude, and other system-specific variables."*⁴⁷ The technique offers an efficient method of control which the user can relate to acoustic phenomena.

[Rodet, 1991; Dodge & Jerse, 1985; Rodet, Potard, & Barriere, 1984; Rodet, 1980]

1.3.5 **Truax**

Granular synthesis is a software instrument associated with three composers, Xenakis, Roads and Truax. In granular synthesis, overall sound is a texture made of many smaller sounds. Control of texture is effected by control of musical parameters at the granular level. Granular synthesis provides a range of techniques for recombining

⁴⁷ Dodge & Jerse, 1985: fonctions d'onde formantique p. 217

and transforming miniature sounds into massed sounds that are capable of great timbral diversity. Truax has developed these techniques over many years. His POD System of Interactive Composition Programs dates back to 1977. There have been extensive updates, one to accommodate the POD-X hardware enhancements, another for a multi-DSP implementation. Truax has used this extensively in his most recent compositions *Riverrun*, *Wave Edge* and *Solar Eclipse*. Granular synthesis techniques also have a special appeal for composers with a background in classic tape studio techniques where the organisation of timbre is a primary compositional focus.

[Garton, 1992; Truax, 1988; Wilde, 1988; Truax, 1985; Truax, 1982; Truax, 1977; Truax, 1979-86: (CD)]

1.3.6 Jaffe

Jaffe is the composer most involved in the musical development of the **Karplus-Strong algorithm**⁴⁸, an algorithm which produces realistic plucked string sounds. Jaffe's *Silicon Valley Breakdown* (1983) is the definitive musical work using this algorithm. The algorithm uses delay lines that are momentarily charged with noise then slowly discharged by averaging adjacent values thereby producing a decay envelope resembling a plucked string. Various modifications to the algorithm have enabled it to produce other more percussive timbres. The attraction of this new software instrument was that it sounded extremely realistic and at the same time was computationally very efficient and simple to realise on 8-bit machines. Moreover, it has provided a starting point for a new musical development known as physical modelling.

⁴⁸ named after the computer scientists who first discovered it

[Smith, 1996; Smith, 1987; Rowe, 1992; Jaffe & Boynton, 1989; Stickney, 1988; Jaffe & Smith, 1983; Jaffe, 1994: (CD)]

New synthesis techniques - or software instruments - have helped computers to find their way into the musical mainstream. The most notable example of this has been Yamaha's synthesiser, the DX7. This was a marriage of VLSI technology of the 80s and a software instrument ⁴⁹ developed by Chowning more than a decade earlier. The DX7 was one of the first software instruments to bridge the divide between computer music research and the music industry. Generic personal computers are now bridging the same divide between current computer music research and today's musician.

The home computer market is constantly introducing musicians to a new generation of software instruments. The sound produced by these instruments can now be downloaded as an audio sample from a remote computer site. Moreover the software which created them can also frequently be downloaded. Many of them are available in the public domain through programs like Csound (see below in 1.4.2 Vercoe). Generic computer technology is rapidly providing the power to implement techniques which provide continuous control of sound at a cost that can now be considered affordable.

⁴⁹ The main software instrument in use on commercial MIDI systems apart from audio sampling has been Chowning FM in the case of Yamaha and other related distortion techniques implemented by Yamaha's competitors. Yet for all the flexibility afforded by musical instrument manufacturers like Yamaha in its implementation of FM, commercial musicians have used factory-presets - demonstration sounds supplied by the manufacturer - as if they were samples. This has been exploited by the industry allowing it to supercede previous models by producing a new model with a further stock of new pre-set sounds.

1.4 Composer-authored Composition Software

The development of new means to play software instruments seems closer to conventional idea of composition than an activity that can be called instrument building. Yet these programs were just as important for the development of the computer as a unique, flexible musical instrument. The systems created display a variety of approaches to composing. These were created by composers conversant with the constraints of computer systems but who devised music composition software to better accommodate their compositional needs.

Initially these programs were written after the **Music4** model for mainframe systems and involved instrument definition as part of the software. The arrival of the microprocessor and then later, the personal computer and MIDI brought greater diversity in new composition software. With MIDI, the software instrument had become a piece of dedicated hardware. The abundance of such hardware made it practical for a growing number of composers to earn their income by designing software for MIDI hardware, with the intention that it is not only used for their own compositions but also by other composers in the commercial field.

Advances in computer science that have taken place since computer music was first introduced at Bell Labs have also introduced a wide variety of programming techniques that have been applied in music composition by composers. Composer/programmers invariably not only program in assembler, Fortran, or C, but Forth, Lisp and a host of other object-oriented languages, and languages designed for DSP processors, multi-processor environments and neural networks. The rapid improvement in computer hardware is being matched by the growing level of expertise among composers who author composition software.

Such composers include :

Joel Chadabe, Phil Burk, Ron Kuivila, Paul Lansky, Richard Moore, Larry Polansky,
David Rosenboom, Robert Rowe, Trevor Wishart, Iannis Xenakis

1.4.1 Chadabe

Chadabe was one of the first composers to write his own software. A program called **Play** was first written in 1977 ⁵⁰. In later composition systems such as **JAM Factory** and **M** ⁵¹ the concept of score includes real-time interaction. The performance is a musical process designed by the composer using software. Compositional decisions can be made by the composer during performance. Chadabe distributed these programs under the his own commercial label ⁵² and his approach to interactive composition thereby indirectly influenced the thinking of many in the field of commercial music.

[Casey, 1992; Chadabe, 1991; Poyourow, 1986; Dodge & Jerse, 1985; Chadabe & Meyers, 1978; Chadabe, 1977; Chadabe, 1984-87: (CD)]

1.4.2 Vercoe

Vercoe developed **Csound** at the Media Laboratory, MIT in 1986. The programs making up Csound have a long history of development in earlier programs by Vercoe called **Music360** (1968), and **Music11** (1973) derived from Matthews' Music4.

Csound is a composition system which incorporates many synthesis modules written by

⁵⁰ Dodge & Jerse, 1985: PLAY II (Chadabe) p. 333 in collaboration with Roger Meyers

⁵¹ written in collaboration with David Zicarelli

⁵² Intelligent Music Systems

other composers ⁵³. Csound source code is available in the public domain. Source files are used to compile digital audio files and provide an ideal means of audio transfer between PC workstations. The 1992 revision of Csound incorporates MIDI converters enabling it to be run from MIDI score-files and external MIDI keyboards. Real-time software synthesis with sensing and control are now a reality. In 1997 Analogue Devices released a DSP-card based on extended Csound ⁵⁴. Vercoe has also developed automated score following algorithms, where the machine follows the rate of progress of the live performance.

[Vercoe, 1997; Vercoe, 1996; Vercoe (Scott) & Boulanger, 1996; Vercoe, 1995; Vercoe, 1992; Vercoe, 1984; Vercoe, 1978: (LP)]

1.4.3 Kuivila

Kuivila ⁵⁵ created a program called **Formula** in 1989. Based on Forth, Formula is a language designed to address problems associated with using continuous musical control - volume, time, timbre etc - in the context of MIDI synthesisers that are essentially event oriented (On-Off). Formula offers a method of envelope shaping and time deformation for continuous functions which can be used with MIDI systems. This reflects Kuivila's early background as a composer building and performing with analogue electronics where controls used to refine the timbral properties of the sound are implemented as time varying cyclic processes.

[Horton, 1991; Reinbolt, 1991; Kuivila & Anderson, 1989]

⁵³ Vercoe, 1992: Csound selected instruments cited at Amsterdam Catalogue of Csound Computer Instruments v1.1 at <http://mars.let.uva.nl/gather/acccci>

⁵⁴ Vercoe, 1997: Extended Csound in Analogue Devices Sharc-cards cited at http://www.analogue.com/publications/press/products/AES_Nxt_092697.html

⁵⁵ with computer scientist David Anderson

1.4.4 Lansky

Lansky was one of several composers who learned hacking as a survival skill in early computer musical environments. Lansky's **Mix** followed by **C-Mix** was a direct-to-disk synthesis and composition system which owes more to Matthews' earlier Music4 rather than the later and more live-performance oriented Groove systems. Like Vercoe's Csound, versions of C-Mix have been ported to Macintosh and IBM PCs.

[Warren, 1993; Belet, 1993; Rowe, 1990; Dodge & Jerse, 1985; Roads, 1983; Lansky, 1997: (CD); Lansky, 1994: (CDs); Lansky, 1992: (CD); Lansky, 1990: (CD); Lansky, 1982: (LP)]

1.4.5 Moore

Moore was one of the co-authors (with Matthews) of the first interactive synthesis language called **Groove** ⁵⁶ for a hybrid digital/analogue system at Bell Labs (1968). Using a keyboard as an input device and its implementation has been compared with a minus one recording where one layer of a texture is played and/or recorded while other textures consisting of more detailed control information recorded in a previous layer can be heard ⁵⁷. This platform was the springboard for many of the later developments in interactive software eg. Play (Chadabe & Meyers) Formula (Kuivila) & Formes (Rodet). Moore's most recent program for software synthesis is CMusic.

[Moore, 1990; Moore, 1988; Dodge & Jerse, 1985; Moore, 1983; Moore, 1982; Moore, 1978]

⁵⁶ main users of the system were composers Emmanuel Ghent and Laurie Spiegel

⁵⁷ Dodge & Jerse, 1985: GROOVE (Moore & Matthews) p. 332

1.4.6

Polansky, Rosenboom & Burk

HMSL - an acronym for Hierarchical Music Specification Language - is a composition language authored by three composers. It is not an application but a general-purpose language aimed at composers who are relatively skilled in programming. Three design criteria are "*extensibility and experimentation, real-time interactive performance and compositional intelligence*" motivated from a desire to "*continually redefine the nature of composition and of performance itself*". Real-time interaction, and collaboration with other artists are some of the musical concerns of Polansky, the main author of HMSL. Most of the pieces eg. *17 Simple Melodies* (1987), *Simple Actions* (1987) involve either other composers, instrumentalists, poets. *B'rey'sheet* (1984) the first piece done by Polansky in HMSL and *Horn* (1990) both use experimental intonation - defined in HMSL and implemented on a Yamaha FB01 synthesiser.

[Polansky, 1994; Burk, Polansky & Marsanyi, 1993; Polansky, Burk & Rosenboom, 1990; Polansky, 1987; Polansky, 1983; Abbott, 1983; Polansky, 1990: (CD); Polansky, 1990: (CD)]

1.4.7

Rowe

Cypher is a Lisp-based interactive computer music system for composition and performance authored by Rowe. It has two main components, a listener and a player. *Flood Gate* (1989) was the first notated piece he wrote using Cypher and is scored for violin, piano and computer. *Rant* (1991) is a similarly notated piece for larger ensemble of seven instruments, with soprano and computer. Whether the music is notated or improvised, the computer characterises what it is hearing and uses that information to generate a musical response. Computer listening involves the development of score following software combined with improvisational algorithms.

[Rowe, 1992]

1.4.8

Wishart

Wishart initially composed music theatre works and works for magnetic tape. A founding member of a British cooperative called the Composer's Desktop Project, Wishart wrote numerous signal processing algorithms for creating and manipulating sound in ways that extend his earlier techniques in tape composition. *Vox* (1982-1989) is a series of works for tape, or tape and voices, which utilise many spectral transformation techniques developed by the composer. Some of these, like **brassage** and **sausage**, are based on granular synthesis; some, like **remix**, involve construction of massive textures by manipulating multiple parameters; others, like **morph**, involve gradual transformation from one sound into another.

[[Wishart, 1998](#); [Wishart, 1994](#); [Wishart, 1988](#); [Wishart, 1973-77](#): (CD); [Wishart, 1982-89](#): (CD)]

1.4.9

Xenakis

Xenakis, architect as well as composer, was one of the first to use computers for generating musical structures. The **Stochastic Music Program** (1978) and the **Gendy** system (1991) are both systems for the application of software algorithms derived from Xenakis' early orchestral works *Metastasis* (1953) *Pithoprakta* (1955) and his electro-acoustic compositions *Orient-Occident* (1963) and *Diamorphosis* (1962). The stochastic music program systems are closely associated with development of the **UPIC** hardware ⁵⁸. This is a CAD system that accepts graphic notation from a light-pen and allows graphic sound editing as part of the compositional process. Where the computer had previously been used to generate individual instrumental parts produced

⁵⁸ acronym for [Unite Polyagogique Informatique du CEMAMu](#)

in the form of graphic plots, the UPIC system translates these graphic plots directly into musical parameters used to play the sound. The system was used in *Mycenes Alpha* (1978).

[Xenakis, 1992; Lohner, 1986; Lohner, 1986; Marino, Raczinski & Serra, 1993; Marino, Raczinski & Serra, 1992; Bernard, 1991; Xenakis, 1982: (CD); Xenakis, 1953-64: (LP); Xenakis, 1957-92: (CD)]

Software devised by these composer/programmers are the tools they needed to produce new musical applications, ie. a composition, a performance using new instruments. The various software tools created by these composers clearly define different ways of composing music. They provide new compositional resources that make it easier for other musicians to work with computers, even if some of these resources tend to impose a particular composer/author's way of thinking on other end users.

1.5 Purpose-built live-electronic instruments & systems post-70's

Since the advent of the microprocessor (1974) and MIDI (1983), the level of activity among composer/instrument-builder activity has become extremely diversified and falls into four categories :

- Automatic Composition Algorithms & Improvising Machines
- Interactive Human Interfaces
- Interactive Hybrid Interfaces
- Interactive Network Interfaces

About the time of the arrival of the first microprocessor, the commercial availability of voltage-control synthesisers gave some musicians direct access to music technology for the first time, although voltage-control synthesiser technology was generally engineered more for recording than live performance ⁵⁹.

Microprocessors played an important role in de-mystifying signal processing concepts, once considered an arcane pursuit for research scientists. Through popular electronics literature, these concepts gradually found their way into the language of musicians building dedicated musical instrument hardware. But it was MIDI which made new techniques for music synthesis accessible for large numbers of musicians, and indeed, the general public. Prior to MIDI, music synthesis had been accessible primarily to those composers who had worked in research institutes and academia.

The introduction of MIDI in 1983 was one of the most important developments in the history of computer music. It not only allowed computer musical instruments made by different instrument manufacturers to be more compatible but involved large numbers of musicians in the development of computer music. This resulted in a gradual merging of two important experimental streams of music : one, the more formal, collaborative effort between computer scientists and musicians, and the other a tradition of experimental composition that had grown out analogue electronics. With MIDI, a greater sharing of musical resources became possible.

⁵⁹ even though the development of new performing instrument technology was involved eg. Subotnick (Buchla synthesiser), Zinovieff, Cary, Banks (Putney VCS-3 synthesiser) or Carlos (Moog synthesiser). Carlos' *Switched-on Bach* (1967) gained widespread acceptance for the musical possibilities of voltage control synthesisers; however, *Switched-on Bach* is more an accomplishment of multi-tracked recording technology rather than the performance capabilities of voltage-control synthesiser technology.

1.5.1 Automatic Composition Algorithms & Improvising Machines

Machines that allowed composers to design their own sound producing algorithms also gave composers an unprecedented ability to define musical detail and this inevitably led composers to automate compositional decision-making processes. The term **algorithmic composition** describes work whose form or a section thereof is generated by a method which rigorously determines the musical detail. It encompasses the work of several composers whose work precedes the advent of the microprocessor.

At first algorithmic composition using computers was purely a process used to generate scores played on conventional instruments. This process that was originally both rigorous and labour intensive as it involved manual transcription of graphic or alpha-numeric code into conventional music notation. It gradually became a method-based means of realising music on a computer in real-time, where any distinction between performing a pre-composed work and improvising is obscured. Microcomputers and ready-made MIDI hardware have allowed algorithmic design to be implemented more easily, more flexibly and with greater sophistication.

As algorithmic composition becomes more interactive it has gradually become absorbed into the language of composers whose focus is still based on traditional models of music production. More composers, recognising the incompleteness of the score as a representation of their artistic intention, have become involved in designing algorithms. Algorithmic composition is no longer simply a process to facilitate the production of scores. It is now also a process for making compositional decisions that govern performance in real-time.

Representative composer/instrument-builders involved are :

Clarence Barlow, John Cage, Warren Burt, Lejaren Hiller, Salvatore Martirano,
Gottfried Michael Koenig, George Lewis, Richard Teitelbaum, Carl Vine, Iannis Xenakis

1.5.1.1 Cage

From the time he composed *Music of Changes* (1951), Cage began to apply chance procedures in composition. Even though it was derived from the I-Ching ⁶⁰, the rigor of Cage's compositional method can be compared to the rigor of algorithmic processes created by other composers using computers. Both involve a means of extending the powers of the composer's imagination, leading to discovery of new musical outcomes at the time of their realisation in performance. It wasn't until *HPSCHD* (1969) that Cage, in collaboration with Hiller, first used computers to implement these algorithms.

[Cage & Tudor, 1960-88: (CD); Cage, 1939-51: (CD); Cage, 1944-45: (LP); Partch & Cage, 1940s: (CD)]

1.5.1.2 Xenakis

Xenakis first implemented mathematical algorithms based on the calculus of probabilities ⁶¹. He used the term **stochastic music** to describe this method of organising large massed sounds. These he employed in his orchestral compositions *Metastasis* (1953) and *Pithoprakta* (1955). Stochastic algorithms used in these works

⁶⁰ J. Cage : 1961 *ibid*

⁶¹ I. Xenakis : Formalised Music publ. Indiana University Press 1970

were devised using a generic computer language ⁶² and realised on a mainframe computer ⁶³ to produce graphic output which was later translated into score. These algorithms were later implemented electro-acoustically in *Mycenae Alpha* (1978) a work which was realised on the first of several purpose-built UPIC systems. Xenakis' GENDY program (1991) developed these ideas into a new synthesis algorithm based on random walks ⁶⁴.

[Marino, Raczinski & Serra, 1993; Xenakis, 1992; Marino, Raczinski & Serra, 1992; Bernard, 1991; Lohner, 1986; Lohner, 1986; Xenakis, 1957-92: (CD); Xenakis, 1978: (CD); Xenakis, 1982: (CD); Xenakis, 1953-64: (LP)]

1.5.1.3 Hiller

Hiller was among the first composers using computers to produce algorithmic compositions. His *Illiac Suite for String Quartet* (1957), done in conjunction with Isaacson, was one of the first musical scores produced by a computer. This generated alpha-numeric code which was later transcribed into musical notation. Hiller ⁶⁵ went on to devise a compositional language called **MUSICOMP** used to implement stochastic choice processes in his *Computer Cantata* (1963), and later *Algorithm I* (1967) and *Algorithm II* (1972) for small chamber ensemble and electro-acoustic tape.

[Hiller, 1981; Hiller & Cage, 1969; Hiller, 1969; Hiller & Baker, 1964; Hiller, 1957: (CD)]

⁶² FORTRAN

⁶³ IBM 7040

⁶⁴ Marino, Raczinski & Serra, 1992 (from CEMAMu studio report with no page numbers)

⁶⁵ in collaboration with Robert Baker

1.5.1.4

Koenig

Koenig implemented two algorithmic composition systems **Project I** (1964) & **Project II** (1967). While Project I was designed to support the production of Koenig's own electro-acoustic composition *Terminus II* (1966), it was Project II which provided a link for other composers interested in aleatoric processes used in the performance of instrumental music. Project II provided composers with control of the tendencies of random choices made by a machine which selected from various musical elements chosen by the composer. Both Project I & II were developed for the dual purpose of controlling voltage-control analogue hardware used in generating electro-acoustic compositions as well as for generating code that could be transcribed into music scores. Koenig's compositions *Terminus II* (1966), *Function Grun* (1967) and *Function Gelb* (1968) were realised on a hybrid machine which produced analogue data for controlling voltage control oscillators and filters to create audio.

[Koenig, 1983; Koenig, 1978; Roads, 1971; Koenig, 1992; Koenig, 1955-69: (LP); Koenig, 1968: (LP); Koenig, 1966-67: (LP)]

1.5.1.5

Martirano

The **Sal-MAR Construction**, a purpose-built instrument designed and built by Martirano ⁶⁶ (1969-72) was one of the first hybrid algorithmic machines used in live interactive performance ⁶⁷. The system uses TTL logic circuits to drive analogue modules, such as voltage-controlled oscillators, amplifiers and filters which in turn drives a matrix of 24 speakers mounted in semi-mobile enclosures.

⁶⁶ in collaboration with engineers Divilbiss, Franco and Borovec at the University of Illinois

⁶⁷ Pellegrino, 1983: photographs of Sal-Mar Construction pp. 95-98

"A changing pattern of sound-traffic by 4 independently controlled programs produces rich timbres that occur as the moving source of sound causes the sound to literally bump into itself in the air thus affecting phase cancellation and addition of the signal. The control panel has 291 touch-sensitive, two-state switches multiplexed to four levels, representing the four programs. The unique characteristic of the switch is that it can be driven manually and logically, which allows human/machine interaction. Most innovative feature of the human/machine interface is that it allows the user to switch from control of macro to micro parameters of the information output ⁶⁸."

Martirano's interest in real-time interaction did not stop with hybrid systems but incorporated compositions for multiple MIDI systems. *Four not Two* (1987) developed on his **yahaSALmaMAC** program (1987-88) ⁶⁹ used a programmable DSP ⁷⁰ to develop time-varying spectra in real-time.

[Rosenboom, 1996; Pellegrino, 1983; Martirano, 1962-92: (CD); Martirano, 1985: (CD)]

1.5.1.6 Burt & Vine

The concept of live-performance algorithms was developed by composers like Burt and Vine who, like Martirano in the 60's, designed and built purpose-built digital circuits to generate both wave-forms and sequences algorithmically. The **aardvarks machine**

⁶⁸ from CD cover notes for *SAL-MAR Construction (Look at the back of my head for a while)* Composer in the Computer Age V Salvatore Martirano Retrospective : 1962-92 Centaur CRC 2266

⁶⁹a system consisting of a Macintosh II computer, a Yamaha music engine, 25 synthesisers and a percussion unit (DX7 (E upgrade), TX8-16, REV7, TX81Zs, RX5), a Zeta violin and a program called SAL (acronym for Sound and Logic) written in Le_Lisp by Sal Martirano and David Tcheng.

⁷⁰ on the KYMA system in collaboration with Carla Scarletti (composer and instrument builder), Kurt Hebel & William Walker in which the Sound and Logic program is translated into Smalltalk 80

designed and built by Burt and the **patent little marvel** designed and built by Vine both used CMOS linear feed-back shift registers where feed-back was controlled by an array of switches operated by the composer during performance. Both designs had multiple and independently-clocked shift registers. In each shift register the outputs were fed through an R-2R resistor summing network to produce a sequence of analogue voltages as the shift register was clocked. The resultant signals were then used either to voltage control the clocks of other shift registers or to produce audio. Such devices which were used in live performances - usually in conjunction with other voltage control equipment - gave rise to a new phase in the development of algorithmic composition in which composer/instrument-builders became involved in improvised performance. Algorithmic composition had become like improvisation which exists only in real-time.

[Burt, 1991; Garvey, 1985; Vine, 1978; Burt & Riddell, 1980s: (CD); Burt, 1974-76: (cas)]

1.5.1.6 Teitelbaum

Teitelbaum, whose roots in live electronic composition dates back to the 60's, is a composer/instrument-builder who has made the transition from involvement with hands-on electronics and improvisation to live algorithmic composition. He was a member of Musica Electronica Viva, a performance group important in the history of development of improvisation using live electronics. His more recent work called *Digital Pianos* (1987) is one in which an instrument interacts with a player by generating sequences of its own from material supplied to the program by the composer prior to each performance. In Teitelbaum's work, composer, performer and instrument builder are one and the same person and his hands-on experience of building electronics hardware has been replaced by the design of software algorithms.

[Rowe, 1992; Rowe, 1992; Rowe, 1988; Teitelbaum, 1985: (CD); Rowe, 1988]

1.5.1.7

Lewis

In the algorithmic compositions of Lewis, sophistication in musical decision-making of the machine algorithm begins to approach the sophistication of those made by the improvising musician. Lewis' acquired skills as a computer programmer (in FORTH) extend his improvising skills playing trombone. His algorithms can be likened to an improvising machine in which a virtual performer makes musical decisions in real-time. The machine is an interactive performance system where the program is influenced rather than controlled by the biological performer. A pitch-to-MIDI converter is used to track the live trombonist. Throughout Lewis' performances the virtual performer has its own recognisable personality and does not adopt any of the style of the live performer. *Voyager* (1992) is part of this single on-going composition project.

[East, 1995; Rowe, 1992; Lewis, 1994: (CD)]

1.5.1.8

Barlow

Barlow's real-time program called **AUTOBUSK** [Barlow, 1990] is a further development of his early work *Colgluotobusisletmesi* for solo piano (1978), in which he addressed compositional problems of interaction between harmonic overtones present in a piano played at various dynamics. This work, though for conventionally tuned piano, was informed by a study of Partch's theories. Barlow's ideas on tuning are discussed in his article Two Essays on Theory [Barlow, 1987]. AUTOBUSK uses control which affects the microtonal and rhythmic behaviour of musical textures. From seed material supplied by the user it generates music according to algorithmic principles. Each instance of the program can realise several voices in real time. AUTOBUSK can be run autonomously or, using more than one computer in series or in parallel. A serial connection causes MIDI output to be interpreted as input control by the next computer in line, whereas a connection in parallel permits synchronised sets of

six, nine or higher multiples of three voices. AUTOBUSK has recently been used to generate sequences from images ⁷¹ to create music to 1930s silent film by Bauhaus artist Oskar Fischinger.

[Barlow, 1998; Rowe, 1990; Barlow, 1990; Barlow, 1987; Kaske, 1985; Abbott, 1982; Barlow, 1980; Barlow, 1978: (CD)]

1.5.1.9 Machover

Machover at MIT Media Labs ⁷² developed Lisp-based pattern-matching algorithms to follow a score in order to create what he calls a **hyperinstrument**. This is an interactive system for one or more instrumentalists in conjunction with computer listening algorithms to monitor what a performer plays. Here "instrument" can mean either a conventional instrument or a MIDI instrument. The system provides virtuoso performers with controllable means of amplifying their own gestures. Machover's *Valis* (1988) is an application of this concept using two MIDI instruments but his *Bug Mudra* (1989-90) best illustrates the concept of hyperinstrument where a system comprising MIDI instruments is coordinated by means of a gestural controller ⁷³ worn by a conductor. Interaction between members of the ensemble is highly synchronised. Players in the ensemble in a sense become part of an instrument.

[Harvey, 1988; Machover, 1987: (CD)]

⁷¹ Barlow, C. image used with AUTOBUSK to generate music to 1930s film by Bauhaus artist Oskar Fischinger, collaborators M. Bischell, B. Wagner at <http://www.ifi.unizh.ch/groups/mml/people/bwagner/barlow/>

⁷² in collaboration with Joe Chung

⁷³ called the dexterous hand master

As computer technology evolved, new algorithms changed the way composers organised musical parameters within a composition. These changes eventually allowed both the performer as well as the machine to be involved in the decision making process at the moment of performance.

1.5.2 Interactive Human Interfaces

Composer/instrument-builders who developed and used alternative gestural devices reflect dissatisfaction among composers with the state of the user interface as it existed in standard musical instrument technology. Redesigning the human performance interface remains the preoccupation of many composers who have attempted to create computer musical instruments whose performance gestures use the whole human body and not just its parts. Here the thrust of technological development is towards a more holistic approach to performance, and includes systems for performance by a non-expert public. Such interactive instruments begin to realise some of the possibilities opened up by electronic technology and have the potential to create new genres of musical performance based on hitherto unused human skills.

Representative composer/instrument-builders involved are :

Joel Chadabe, Stuart Favilla, Rolf Gehlhaar, David Rosenboom, Stelarc, Michel Waiswicz

1.5.2.1

Chadabe

Chadabe in *Solo* (1980) ⁷⁴ used movement gestures as an alternative method of control instead of using a standard music keyboard. The instrument controlled was the Synclavier ⁷⁵. Hand movement gestures were monitored using a Theremin, one of the earliest electronic instruments, which use antennae to translate into musical information the presence of the human body in an electromagnetic field ⁷⁶. Movement gestures were used to influence the stochastic generation of pitch, rhythm, timbre and texture. Though made with the arms and hands, the gestures are essentially choreographic focusing attention on the action of the body more than the machine.

[Casey, 1992; Chadabe, 1991; Poyourow, 1986; Dodge & Jerse, 1985; Chadabe & Meyers, 1978; Chadabe, 1977; Chadabe, 1984-87: (CD)]

1.5.2.2

Favilla

Favilla has designed several instruments where laser light forms the virtual strings of an instrument. Though it is played by the movements of the hands - in the manner of a harpist - it also has the potential to be configured to be played by the human body. The earliest of these instruments called an **LDR Controller** ⁷⁷, **Light-harp** ⁷⁸ and **LaserLyre** ⁷⁹ link physical movements with musical gestures such as monophonic pitch

⁷⁴ Dodge & Jerse, 1985: *Solo* (Chadabe) p. 332

⁷⁵ New England Digital Synthesiser

⁷⁶ This instrument had also been adapted for use in dance most notably by Tudor and Mumma to interface dance movement as a controlling device for musical gesture [see Tudor, Mumma, Cage's *Variations IV*] and in the electronic music and dance projects done between 1970-75 by choreographer Philippa Cullen [see Alexander, 1989; Cullen, 1989 & Wallace, 1989].

⁷⁷ in collaboration with Robin Whittle

⁷⁸ Favilla, 1998: Light Harp photograph cited at <http://www.music.utas.edu.au/favilla/IPS.html>

⁷⁹ both Light-harp and LaserLyre built in collaboration with David Brown

inflections found in non-western music ⁸⁰. The third of these has been used in his composition *Alapana* (1993). Composition and performance of polyphonic music in non-equally tempered tuning systems is the focus of Favilla's extended MIDI-based design aimed at maximising gestural feed-back and developing performer skills.

[Favilla, 1998; Favilla, 1996; Favilla, 1994; Favilla, 1994; Favilla, 1994: (CD)]

1.5.2.3 Gehlhaar

A similar use of performer's spatial movements as a means of musical control has been a long time preoccupation of Gehlhaar, a composer/performer who worked with Stockhausen in many of that composer's live-electronic works. Since the early 70s, Gehlhaar was closely involved with interactive sound installations which led to the design of an interface for monitoring movement using ultra-sound. Funded by IRCAM in 1984, the system, which Gehlhaar calls **SOUND=SPACE**, mapped a rectangular space using an array of ultra-sonic ranging devices. Various topographies were developed which allow various zones of the instrument either to make the music or influence the musical outcome. The musical applications of Gehlhaar's SOUND=SPACE project include "*an environment open to the general public, as a creative resource for both the physically and mentally handicapped, as an "instrument" for dancers to create their own musical accompaniment and as a musical instrument for performance in concert, either solo or in the context of an ensemble*" ⁸¹.

[Gehlhaar, 1991]

⁸⁰ notably Carnatic (south Indian classical) and Vietnamese music

⁸¹ Gehlhaar, R. [1991]: p.71

1.5.2.4

Rosenboom

Throughout the 70's work on biologically adaptive instruments was done by musicians continuing in the footsteps of Lucier, most notable among them being Teitelbaum and Rosenboom. *Portable Gold & Philosopher's Stones* (1972), Rosenboom's first work using biofeedback monitoring to track and process EEG potentials of four biofeedback musicians allowing them to improvise musical parameters which determined the spectral composition of the music. *On Being Invisible* (1976-1979), his major work of that period, was an on-going performance project incorporating performances, which he describes as "*a self-organising, dynamical system, not a fixed musical composition.*"⁸² This is an interactive instrument where the distinction between the technology and the biological performer becomes blurred. The instrument, or system, functions in a semi-stable state to produce musical patterns which may evolve as the result of the two way interaction between the machine and the biologically-monitored performer.

[Rosenboom, 1996; Belet, 1993; Rosenboom, 1990; Rowe, 1991; Abbott, 1983;
Rosenboom & Braxton, 1992: (CD)]

1.5.2.5

Stelarc

The concept of biologically adaptive instrument has been taken a step further in the 1980s by performance artist Stelarc. In his on-going *Third Hand* (1987-) performance project, biological monitoring is used as part of a live performance event to control a large industrial robotic arm as well as live electroacoustic accompaniment⁸³. Stelarc's

⁸² Rosenboom, D. [1990]: p. 51

⁸³ produced with various sound collaborators - Densil Cabrera (1989), Rainer Linz (since 1991)

body performance-instrument was further extended in two recent on-line web performance/installations. *Ping Body* (1996) and *Parasite* (1997) involve web interaction between the artist and members of the public.

[Stelarc, 1996-7; Stelarc, 1992; de Marinis & Stelarc, 1992; Burt, 1991; Stelarc, 1988]

1.5.2.6

Waiswicz

At Steim ⁸⁴ in Holland much work has been done in the field of alternative gestural devices. Set up by Waiswicz, this work has led to systematic development of hardware called the Sensor-lab which is designed not only to support Waiswicz's own interactive performance projects eg. *The Hands* (1985), and *Touch-Monkeys* (1987), but is intended as an open-ended interface that can be used by any of the various performers and musicians who have worked at Steim. The sensor-lab is a microcomputer platform for interfacing a variety of tactile sensing and other monitoring devices to MIDI. Such monitoring devices include the Web, an extension of the Theremin with a large antenna, which like its predecessors, was developed in collaboration with dance. Waiswicz's concept of interactive musical instrument came from exploring the medium itself rather than trying to re-build conventional musical instruments using electronic technology. Instead, direct hand contact is used to change the resistance as well as capacitance of an electronic circuit. These changes are mapped by means of a microcomputer into MIDI control information.

[STEIM, 1998; Waiswicz, 1993; Ryan, 1992; Rowe, 1990; Lehrman, 1986; Waisvisz, 1984; Waisvisz, 1998]

⁸⁴ at a collaborative arts institution set up by Michel Waiswicz

New gestural devices that offer an alternative control medium to performance devices such as keyboards or control devices based on other standard non-electronic instruments, have opened pathways into other artistic disciplines helping to redefine the nature of musical performance.

1.5.3 Interactive Hybrid Technology Interfaces

The term **interactive hybrid technology** is used here to describe an interface that deliberately forms musical links between interactive computer technology and a traditional musical instrument or some other medium not directly related to music. Composers making such links between the two technologies usually revise the concept of musical performance, emphasising the composer as performer, or, in some cases the audience as participant. The hybrid interface most commonly evokes a new situation for a listener by placing a recognisable musical instrument in an evocative and sometimes bizarre visual or extra-musical context. Such musical instrument design is necessary to highlight facets of musical presentation that have become obscured by the historical development of musical genres which evolved in parallel with musical instrument technology from an earlier epoch.

Representative composer/instrument-builders involved are :

Ben Neill , Rod Berry, Joel Ryan, Jerry Hunt, Alistair Riddell, Nicholas Collins, Jacques Dudon, Godfried-Willem Raes, Paul de Marinis

1.5.3.1 Neill

Neill, a classically-trained composer/performer, invented the **Mutantrumpet** to extend the timbral style of his playing using both acoustic and electronic means. This triple-

belled instrument, tuned in just intonation, allows rapid transformations of acoustic timbre as well as various combinations of muting and straight playing to be used simultaneously. In *678 Streams* (1993) for mutantrumpet and computer interactive electronics "each of the three bells of the mutantrumpet is responsible for one of the three sets of harmonics" ⁸⁵. His compositions usually involve multimedia controlled through live interaction with his instrument. *Downwind* (1992) is based on tapes and transcripts of residents of a town in Ohio who sued the government over nuclear plant contamination while *ITSOFOMO* (1993) is a multi-media collaboration with an artist, writer & activist who died of AIDS; and *Greenmachine* (1994) a multi-media installation/performance.

[[Reinbolt, 1991a](#); [Reinbolt, 1991b](#); [Neill, 1996](#): (CD); [Neill, 1993](#): (CD)]

1.5.3.2 Berry

Hack'n'hocket (1991) and *Rotary Zithers* (1991) are two performances by Berry. Both works use three sound-sculptures, also called **rotary zithers**. Each zither has up to thirty strings spanning a range of about four octaves. Each zither mounted in an upright position revolves at about 120 rpm while being played by plectra driven from a stationary platform. Purpose-built hardware allows MIDI commands to drive solenoids which activate several plectra on each zither. Which strings are plucked is indeterminate although timing of events is reasonably precise. Plectra are fired arbitrarily using an obsolete MIDI sequencer, in which MIDI is generated isorhythmically from separate pitch and rhythm tables. Berry has used similar systems where both wind and stringed instruments are played by solar power and activated in different conditions of sun-light in *(i heard) YOUR FOOTSTEPS* (1995) and *(i felt)*

⁸⁵ from CD liner notes [Neill, 1993](#)

YOUR BREATH (1995). Audience moving in the vicinity of the panels also affect the operation of the solar panels which control the system.

[[Berry, 1998](#)]

1.5.3.3 **Ryan**

Ryan, working at Steim. Joel Ryan has developed ⁸⁶ an interactive media art work *LINA* (1985) in which the behaviour of cellular automata was interpreted in real-time as a means for control of both musical and graphical processes. Ryan has also collaborated with commercial development of an instrument for spatialising sound in his piece called *The Number Readers* (1988). The project involved development of a program to modulate the motion of sound sources using a system of patched functions which allowed the composer to select patterns of modulation as part of the performance ⁸⁷.

[[Ryan, 1992](#); [STEIM, 1998](#)]

1.5.3.4 **Hunt**

From the time of his earliest reported work in *Source Magazine* (1967), Hunt has recognised the role of memory in musical composition and the role of the performer as part of a retrieval system. Hunt's on-going performance project *Haramand Plane* (1971) parallels the development of a live-performance instrument based on performer-activated play-back of sound samples from a cartridge recorder. Hunt ⁸⁸ developed a

⁸⁶ in collaboration with visual artist Ray Edgar

⁸⁷ STEIM [1998]: information about Joel Ryan at <http://www.xs4all.nl/~steim/main.html>

⁸⁸ assisted by David McManaway

tone-based time code system for retrieval of sounds from the cartridges in real-time performance. Cartridges sometimes play back pre-recorded material, sometimes re-processed samples from the same live-performance using highly theatrical performance gestures to access the sampled sounds. The performer is part of an elaborate retrieval system consisting of multiple mechanical tape-transports. Hunt's performance exploits the limitations of these sequential access memory devices. In *Ground : Five Mechanic Convention Streams* (1993), performance control of mechanical tape-transports was activated by arrays of mechanical and electro-mechanical switches mounted in assorted purpose-built sculptured objects. These are played by Hunt during performance. The performance gestures sometimes incorporated interactive video.

[Hunt, 1991; Monaghan, 1987; Hunt, 1967; Hunt, 1993: (CD); Hunt, 1988: (CD); Hunt, 1977: (CD)]

1.5.3.5 Riddell

In 1982, after experiments to create computer files of FSK data to drive an early pre-MIDI computerised player piano, Riddell created several works sequenced⁸⁹ rather than played from the keyboard. These included *Atlantic Fears* (1983) and *Variations for 2 Player Pianos* (1983). He devised⁹⁰ an automated solenoid-driven control action to replace the conventional mechanical action of the player piano in early 1984⁹¹ featuring a new automated strike-action capable of being driven at low audio rates such that the hammers produce FM-like timbres from the strings. The ability to vary the energy of solenoid actuation so that the striker could be held in contact with the string

⁸⁹ after the manner of Nancarrow's player piano compositions

⁹⁰ in collaboration with Harry Tegar

⁹¹ Riddell, A. [1998]: *Atlantic Fears & Black Moon Assails* (& others), with photographs of modified computer-controlled piano action, photographs cited at <http://alphalink.com.au/~amr/ccp/>

after the string was initially struck meant that it was possible to produce pitch shifts coloured by subtle timbral changes. Independent dynamic control over the action of hammers gave the new instrument the capability to create different timbres using piano harmonics. These techniques are used in *Black Moon Assails* (1987).

[Riddell, 1998; Burt, 1991: pp. 164-165; Riddell, 1988; Burt, 1983: pp. 38-40; Burt & Riddell, 1980s: (CD); Riddell, 1998]: (CD)]

1.5.3.6 Dudon

Dudon uses patterned glass disks to generate music. By shining a light through a spinning disk onto a photoreceptor, Dudon is able to produce periodic waveforms which become audible as timbres. By moving a slotted scrim between the light source and disk, and by moving from the inner to the outer section of the disk & vice versa, he is able to produce series of pitches. Evanescent and organic-sounding fractal timbres are generated using as many as 500 slotted disks. Some disks are used as sequencer disks at a slow rate of rotation, and other disks as timbral disks which rotate much faster. By using 6 or 7 different photodiodes and rotating disks, along with volume pedals to switch between the different timbres/sequences, Dudon can produce sonic tapestries of remarkable subtlety and sophistication ⁹². Works like *Fleurs de Lumiere*, *Tournesois* & *Surakarta* music have strong poetic association with nature. Their tuning by virtue of the photo-mechanical processes inherent in the instrument design is entirely in just intonation.

[Dudon, 1996; Dudon, 1994; Dudon, 1987; Dudon, 1986; Dudon, 1987; Dudon, 1996: (CD)]

⁹² Dudon, 1996: description of photosonic disks system in *Lumieres Audibles* on Dudon home page at <http://w3.teaser.fr/~daschour/dudon.html>

Collins, created his first work *Pea Soup* (1975) consisting of elaborate feed-back circuits using analogue electronics. Collins' first digital instruments ⁹³ required group interaction and were specifically designed not to be played by a single player in *Little Spiders* (1981). He has continued to work with purpose-built computer circuitry. This has been used either to create a live mix from pre-recorded sound sources as in *Is She/He Really Going Out With Him/Her/Them?* (1982) and *The Devil's Music* (1985). In these works microscopic samples are appropriated and rearranged from multiple simultaneous radio broadcasts in order to create a live remix for broadcast on a single frequency. Collins has also devised a performing instrument consisting of a modified digital reverberation unit and/or acoustic feed-back controlled from the mechanics of a Trombone in *Tobabo Fonio* (1986-) ⁹⁴, an on-going performance project first performed 1986. Collins' recent creation, the **MIDI Concertina** built at Steim in 1988 ⁹⁵, is a marriage of technology designed by Charles Wheatstone, live-electronics feed-back after the manner of Lucier ⁹⁶, and MIDI electronics.

[Collins, 1993; Collins, 1991a, 1991b; Reinbolt, 1991a, 1991b; Suzuki, 1988; Pslek, 1986; Collins, 1985: (CD)]

⁹³ after an idea derived from Wolff's "co-ordinations" between groups of players

⁹⁴ in progress

⁹⁵ with the help of Paul Spaanderman, Peter Cost and Tom Demeyer

⁹⁶ his composition teacher

1.5.3.8

Raes

Raes' work probably fits into a number of categories ⁹⁷, but hybrid technology seems to feature in most of his work. His most recent work *Sotto Voce* (1991) is the product of his long-standing involvement with sound-sculpture, music-theatre and collaborative performance. Raes in collaboration with members of the Belgian-based Logos ensemble developed interactive instruments such as the **Singing Bikes** and an ultra-sonic Doppler radar. Sound sculptures combine electro-mechanical and semiconductor electronics, both digital and analogue, and include the **Synthelog**, **Bellenorgel**, **Beltelefon** and **Beltotem**. The last of these sculptures interact with the general public by making electronic/electro-mechanical sounds in response to vocal sounds. In *Hex* (1987) amplified electro-mechanical sound producing devices interact with feed-back circuits activated under microprocessor control.

[Raes, 1997a, 1997b; Raes, 1992]

1.5.3.9

de Marinis

de Marinis's sound composition is distinguished by a uniqueness of presentation matched by diverse technical skills ranging from writing DSP software to constructing purpose-built electronics. The range of presentation formats include live performance as well as interactive sound installations some of which have been exhibited at the Exploratorium in San Francisco ⁹⁸. In *The Edison Effect* (1991)⁹⁹, purpose-built 400mW lasers are used to scan and play-back audio data pre-recorded on an

⁹⁷ Raes, 1997]: Holosound installation, Ars Electronica, Linz cited at <http://www.ping.be/logos/holosound.html>

⁹⁸ Behrman, 1991: p. 139

⁹⁹ de Marinis, 1991: *Fragments of Jericho* from *The Edison Effect* at <http://www.babelweb.org/virtualistes/galerie/jerichoa.htm>

assortment of mechanical surfaces; a gold-fish swimming through the beam plays snippets of a recorded popular classic (the *Blue Danube Waltz*). *Mechanisation Takes Command* (1992) is a performance in which a dataglove is used to control a Texas Instrument voice synthesis chip programmed by the composer. *Chaotic Jump-rope* (1993) is a performance installation in which a stretched length of rope is actuated at both ends by two DC motors to produce transverse waves visible to both the audience and performer. By touching the stretched rope, the performer produces chaotic changes in the transverse wave pattern. Purpose-built hardware translates these changes into electrical impulses, which are mapped into MIDI information, and made audible using Max software.

[de Marinis, 1993; de Marinis & Stelarc, 1992; de Marinis, 1991; Behrman, 1991; de Marinis, 1985-86: (CD)]

Interactive Hybrid Technology provides a fresh way to articulate musical sounds which might have become standard through over use or commercialisation and mass-production. A significant feature of hybrid interfaces is the way old technology can be recycled to create new instruments, a bi-product of the rapid rate of development that has taken place in electronic technology. The purpose-built nature of these hardware and software instruments has given composers a new sense of independence as well as giving the musical performances a new sense of vitality. These composer/instrument-builders have adapted and recycled mass-produced technology to suit their own ends. And with the rate at which new technology becomes obsolete, there is every reason to believe this linking of different technologies will continue to be a feature of musical creativity.

1.5.4 Interactive Network Interfaces

Shortly before the arrival of MIDI, groups of composers played with the possibility of making several microcomputers communicate with one another during performance and explored ideas of group performance which grew out of the new capability for machine-to-machine interaction. These machines were not only musical instruments that could improvise according to a composer's algorithmic design or offer unprecedented possibilities for interfacing between the physical world and the imaginary world. These were instruments that allowed a mode of ensemble performance where a new type of social interaction was possible between musical collaborators. The shared musical space was no longer simply a local acoustic space but a virtual environment that was potentially global. MIDI helped to make this idea enter the mainstream by offering a communications protocol that was simple and easy to use. Eventually the need for an improved musical instrument protocol based on generic communications hardware and global musical perspective grew out of dissatisfaction with MIDI's limitations¹⁰⁰.

Representative composer/instrument-builders involved are :

John Bischoff, Phil Harmonic, Rich Gold, Jim Horton, Carl Stone, Mark Trayle, Chris Brown, Scot Gresham-Lancaster, Tim Perkis, Jean Piche, Osamu Shoji, Martin Wesley-Smith, David Wessell

¹⁰⁰ Moore, R. [1983], Loy, G. [1985], Scholz, C. [1991].

Possibly the first simultaneous network performance was by Bischoff, Gold ¹⁰¹ and Horton in 1978. It predates the arrival of MIDI. In this work and performances by other groups like the League of Automatic Music and later, the HUB, there is a sense of optimism that music based on such technology would rely on group interaction and would *"free the computer musician from the mores of highly-structured business and academic institutions."* ¹⁰²

Some of the first documented descriptions of these interactive performances indicate that three compatible microcomputers produced sound and at the same time were used to send frequency related data to each other in order to influence decision making in the other machines in the network. *"It was John Cage who pioneered an important form of collaborative music, that is of the simultaneous playing of compositions. An extension of that idea is to write "reactive" compositions that can interact with one another as well as with their players. This approach makes possible a collective style of music while allowing each composer the opportunity to invent and play complete designs not necessarily subordinated to other parts or wholes."* ¹⁰³

Bischoff, 1991; Bischoff, Gold & Horton, 1978; Horton, 1996: (CD)]

¹⁰¹ One can also expect to see these ideas emerge in the mainstream of future technological development; Rich Gold currently works in the research team for ubiquitous computing at the Rank Xerox centre at the Palo Alto.

¹⁰² Bischoff, Gold & Horton [1978] p.28.

¹⁰³ *ibid* p.28.

1.5.4.2 **The League of Automatic Music : Bischoff, Harmonic, Gold, Horton, Stone, Trayle**

Groups based in San Francisco known as the League of Automatic Music, influenced by the ideas of Bischoff, projected communality rather than a traditional hierarchical value system where a composer at the top of the pyramid directs performers further down the chain of command.

[Bischoff, 1991; Reinbolt, 1991; Wold, 1991; Bischoff, Gold & Horton, 1978; Horton, 1996: (CD); Trayle, 1991; Sirota, 1991; Hub, 1993: (CD); Hub, 1991: (CD)]

1.5.4.3 **The Hub : Bischoff, Brown, Gresham-Lancaster, Perkis, Trayle**

The group called the Hub evolved from the League of Automatic Music. By the time the group had become known as the Hub, the network topology had evolved somewhat from a ring to a star configuration, no doubt influenced by the decision to include a much wider variety of processors now operating in the network. Members of the HUB build their own instruments (acoustic and electro-acoustic) and design their own software. The HUB also uses an assortment of commercial synthesisers as well as voltage control machines and purpose-built interfaces built from assorted, usually recycled, musical instrument technology. *"The term "HUB" actually refers to an interactive program written largely in Assembly language, run in ROM on a centralised computer. The purpose of the HUB-box is to send and receive messages from the other composer/performer workstations."* ¹⁰⁴

[Trayle, 1991; Sirota, 1991; Schrader, 1991; Hub, 1993: (CD); Bischoff & Perkis, 1993: (CD); Hub, 1991: (CD)]

¹⁰⁴ Schrader, B. [1991] p.96

1.5.4.3

Digicon *Night satellite* project : Piche, Shoji, Wesley-Smith

Collaborative composition was a feature of the Digicon *Night satellite* project (1983)¹⁰⁵, which was probably one of the first attempts to link performance in remote locations using satellite technology, though this link-up was used only to transfer music between systems rather than musical control data. Music was bounced via satellite between simultaneous performance using three Fairlight CMIs playing in separate interactive compositions composed and played respectively by Piche (Vancouver), Shoji (Tokyo), and Wesley-Smith (Sydney). In addition to the musical data transferred between earth-stations the link also carried a private line which was used for communication between performers. In addition to performances heard from the other two-earth stations, the satellite link between the three earth stations allowed each performer to hear a delayed version of his own performance, the total delay being the sum of transmission and re-transmission times to and from the satellite. Though *Night-Satellite* does not use a network in the strictest sense it dealt with related problems of synchronisation that would be present in a wide-area musical instrument performance. Piche is also one of the creators of **Cecilia**¹⁰⁶ and **Cynthia**, two graphic interface programs for Csound written in tk/tcl.

[Piche, 1998; Kendall, 1982: (LP)]

1.5.4.4

Wessell

Wessell was part of the commercial and academic collaborative research team which formulated the ZIPI specification¹⁰⁷. Wessell is a composer whose interests include

¹⁰⁵ Piche, 1983: cited at <http://mistral.ERE.UMontreal.CA:80/~pichej/mus.html>

¹⁰⁶ Cecilia, an SGI program, was awarded an equal first prize for computer music software at Bourges in 1997.

¹⁰⁷ first announced in September 1994

musical control of timbre and the application of neural networks in recognition of musical gesture from MIDI keyboards, guitar controllers and a range of other sensing devices ¹⁰⁸. These interests have led to the adaptation of standard communications network hardware for musical instruments. Wessell was one of the co-authors of the specification for ZIPI, an instrument protocol called MPDL ¹⁰⁹. This was the first musical instrument network based on the OSI model of network protocols, and was seen as a replacement for MIDI.

[McMillen, Wessell & Wright, 1994a, 1994b; McMillen, Simon & Wright, 1994; Rowe, 1992; Wessell, 1991; Wessell, 1979; Wessell, 1978: (CD)]

The potential for new types of interaction between groups of people is being explored by composers through network technology. These composers have contributed to the development of virtual instruments which extend the spatial domain of the user. Music in the future will involve the use of shared space and resources in its creation.

¹⁰⁸ Rowe, 1992: pp.87-88

¹⁰⁹ acronym for Musical Parameter Definition Language

1.6 Conclusion

This survey shows the extent to which composers have played an active part in the development of musical instruments during the latter half of the C20th. In past musical epochs, composers whose works have had a major impact on subsequent developments in the history of music have often been closely involved with the design of new musical instruments. In the latter half of the 20th century, most forms of work have come to require a range of skills. As this survey has demonstrated, a new breed of multi-skilled composer/instrument-builder has gradually emerged where composer and instrument-builder are one and the same person. Electronic technology has made it possible for creative musicians to transcend the segmentation of musical roles, eg. composer, performer, instrument-designer, patron, producer, distributor, and audience that was predominant in 19th century. With the arrival of semiconductor technology creative musicians found another means to produce their music and make it available. Electronic technology has provided musical possibilities that had never existed previously.

In the next chapters, the author's interaction with the rapidly evolving musical instrument technology known as MIDI is examined. Particular emphasis is placed on how purpose-built hardware has developed as the result of several musical compositions. In these compositions, the author first realises some of the musical possibilities that have become available through MIDI. In the process, these compositions - discussed in Chapter 2 - help to define the initial specification for a new interactive musical instrument design - described in the main body of this thesis. In the work of the author - like other composer/instrument-builders reviewed in this chapter - composing and instrument-building are inseparable.